Search for type III seesaw heavy leptons in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector

Master's Thesis Porposal

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Motivation

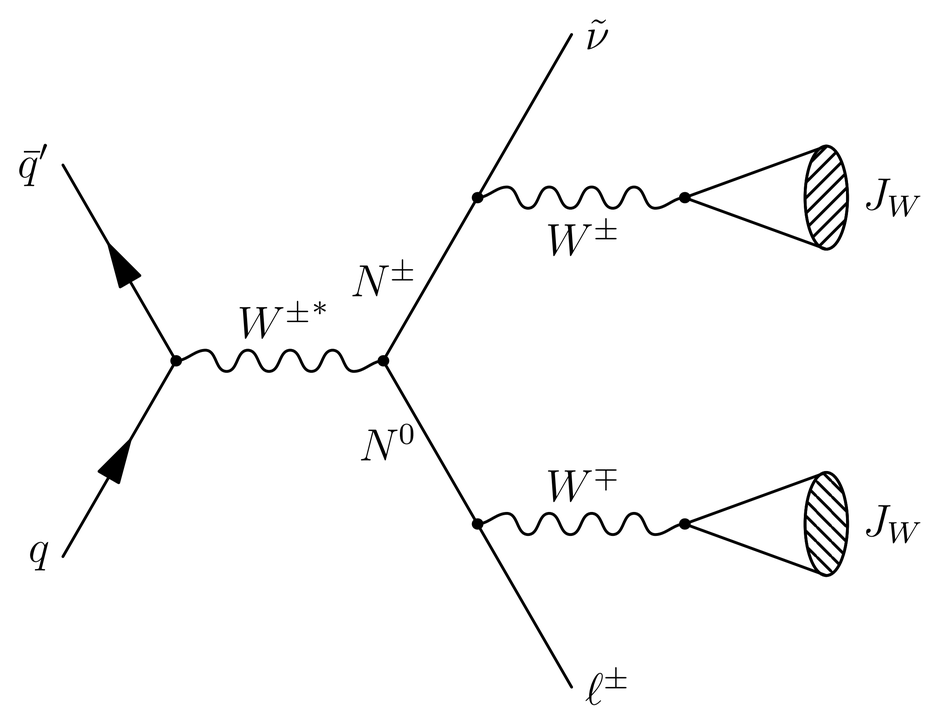
The Standard Model (SM) of particle physics, despite its remarkable success in describing fundamental particles and their interactions, fails to explain the observed non-zero neutrino masses. Neutrino oscillation experiments have conclusively demonstrated that neutrinos have mass, yet the SM predicts them to be massless. This discrepancy represents one of the most compelling pieces of evidence for physics beyond the Standard Model. Several theoretical frameworks have been proposed to explain the smallness of neutrino masses, with the seesaw mechanism being among the most elegant and well-motivated. The **type III seesaw (T3S) model** extends the SM by an $\mathrm{SU}(2)_L$ triplet of fermions that transform as $(\mathbf{1}, \mathbf{3}, 0)$ under the SM guage group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, commonly refereed to as heavy lepton triplets. These triplets couple to the SM leptons and the Higgs field, generating neutrino mass via the seesaw mechanism after the electroweak symmetry breaking (EWSB):

$$L_\Sigma = i\left(\overline{\Sigma} D \Sigma\right) - \frac{1}{2}\left(\overline{\Sigma}^c M_\Sigma \Sigma + \overline{\Sigma} M_\Sigma^\dagger \Sigma^c\right) - \sqrt{2} \sum_{\alpha=e,\mu,\tau} \left(\overline{L}_\alpha Y_{\Sigma\alpha} \Sigma \widetilde{\Phi} + \widetilde{\Phi}^\dagger \overline{\Sigma} Y_{\Sigma\alpha}^\dagger L_\alpha\right),$$

For **TeV scale triplet fermions**, not only the extremely small pair production cross section of the triplet pairs further suppresses the sensitivity of the multileptonic signals but also the small branching fractions of the leptonically decaying SM bosons and the relatively larger SM backgrounds. Hence, to improve the sensitivity one has to consider the hadronic final states produced by hadronically decaying SM bosons, which are known to be embedded with large SM background contributions. The characteristic signature of TeV scale triplets at the LHC would be vastly boosted final state lepton and SM boson that are likely to be largely separated in the azimuth plane. For the daughter SM bosons boosted to large p_Ts, their hadronic decay and fragmentation yield a collimated spray of hadrons which a standard jet algorithm would reconstruct as a single jet and thus are more likely to manifest as a single large-R jet rather than two resolved jets.

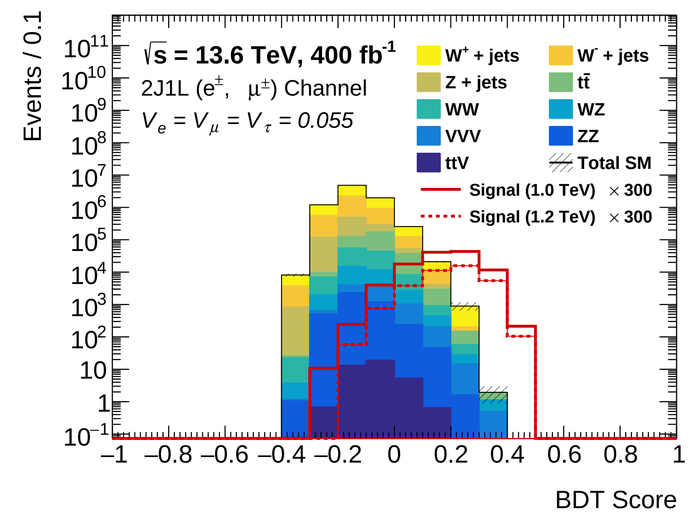
In this analysis, we consider final states that include multiple large-R jets and charged leptons. We use novel techniques for tagging the signal bosonic large-R jets and multivariate techniques to classify our signal events from SM background events. Results from every channel are then combined together and used to set expected exclusion limits on the triplet lepton masses using 400 and 140 fb^{-1} of pp collision luminosity.

Feynman Diagram of the 1L2J Channel

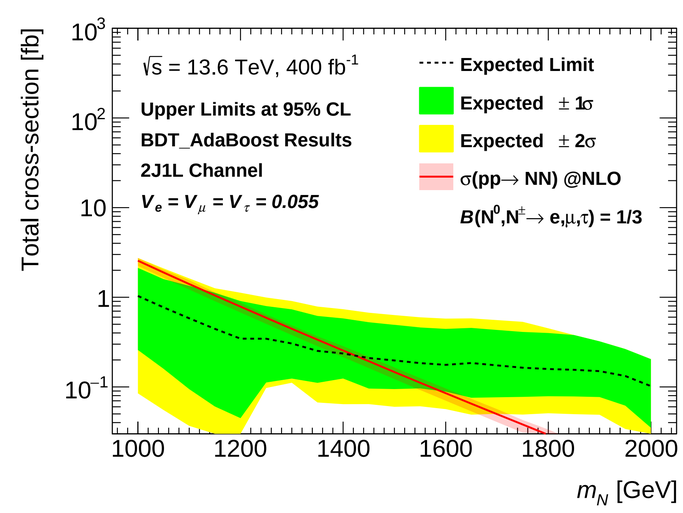


Methdology and Primary Results

We employ the BDT (AdaBoost) method from the TMVA package within the ROOT analysis framework, trained with 27 discrminative variables describing both objet and event kinematics.



We use the pyhf python package for constructing the binned-likelihood model. Upper limits on are obtained using the CL_s method, for $CL_s = 0.05$ (95% CL).



Summary and Outlook

- **Targets ATLAS Run-3 data ($\sqrt{s} = 13.6$ TeV).** Extends previous searches performed at s=13 TeV and below by targeting the higher center-of-mass energy and larger Run-3 dataset, thereby increasing potential signal yield and discovery reach.
- **Comprehensive multivariate analysis pipeline.** Rather than a single classifier, the analysis integrates multiple MVAs: constituent-level ML taggers for boosted boson identification and boosted decision trees for event-level signal discrimination. This layered MVA design lets each stage learn complementary features and reduces model reliance on any single observable.
- **New discriminative variables for Type-III signatures.** We propose and test novel combinations of jet substructure observables (including N-subjettiness ratios and energy-correlation function derivatives) specifically optimized to separate the expected decay topologies of Type-III fermion triplets from dominant backgrounds.
- **Constituent-level ML for fat-jet tagging.** Strong classifiers trained on particle-flow (constituent) inputs improve tagging performance for W/Z/H fat jets relative to taggers based solely on high-level variables. Their outputs are fed into the event-level BDTs to amplify discriminating power.
- **Synergy of substructure observables and ML outputs.** By incorporating both engineered substructure features and ML-derived tagger scores into the final BDT, the analysis exploits orthogonal information channels (physics-motivated observables + data-driven classifiers) to maximize sensitivity

KEY REFERENCES

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[2] **CMS-2012** (10.1016/j.physletb.2012.10.070), **ATLAS-2015/8** (10.1103/PhysRevD.92.032001), **ATLAS-2015/9** (10.1007/JHEP09(2015)108), **CMS-2017** (10.1103/PhysRevLett.119.221802), **CMS-2020** (10.1007/JHEP03(2020)051), **ATLAS-2021** (10.1140/epjc/s10052-021-08929-9), **ATLAS-2022** (10.1140/epjc/s10052-022-10785-0), **CMS-2022** (10.1103/PhysRevD.105.112007)

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